Chapter 4 - Combinational logic

Design circuit

1. Determine number of required inputs & outputs.
2. Derive truth table.
3. Obtain simplified Boolean functions.
4. Draw logic diagram & verify correctness.

Decoder

- Combinational circuit that converts binary info from n input lines to a maximum of 2^n unique output lines.
  → n-to-2^n decoder
- can be a minterm generator

2-4 decoder

2 inputs, 4 outputs

S1	S0	Q0	Q1	Q2	Q3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

enable inputs

- in order to activate & deactivate the device
- EN=1 (is on so will behave normally)
- EN=0 (is off so nothing ever happens)

3-8 line decoder

- 3 inputs, 8 outputs

Inputs			Outputs							
x	y	z	D0	D1	D2	D3	D4	D5	D6	D7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

active-high decoders → generate minterms

active-low decoders → generate max terms

Encoder

- performs the inverse operation of a decoder
- has 2^n (or fewer) inputs and n outputs
- the output lines, as an aggregate, generate the binary code corresponding to each input value.

Ex. 8-to-3 encoder (8 inputs, 3 outputs).

- ambiguity: only one input can be active at any given time or else weird stuff happens.

Decoder expansions

- larger decoders can be constructed using a number of smaller ones.
  → Hierarchical design
- Ex: a 4-to-16 decoder can be designed using five 2-to-4 decoders. (you can use one decoder as an enabler for the other 4 decoders.)

Priority encoder

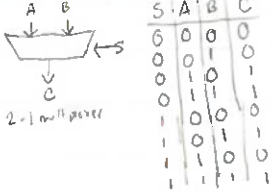
- if 2 or more inputs are equal to 1 at the same time, the input in the highest-numbered position will have precedence.
- valid out but indicated, is set to 1 only when one or more inputs are equal to 1.
  V= OR opic of all inputs

Multiplexer (MUX)

- Selects one of several input signals and passes it on to the input.
- Routing of selected data input to output is controlled by select inputs



- Combination circuit with 2^n data inputs, 1 data output and a number of bit control input that select one of the data inputs



S	A	B	C
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

C = S'AB' + S'AB + SAB
= SA'B + SAB
= S'A(B'+B) + SB(A'+A)
= S'A + SB

- rule to follow:
  2^n data inputs, 1 data output, n selectors